

# SHETH L.U.J AND SIR M.V COLLEGE

## SUBJECT : PYTHON FOR DATA SCIENCE

### 1. Title of Project

#### Tariff Impact Analysis on Product Prices Using Python

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### 2. Introduction

Tariffs are taxes imposed on imported goods and can affect the final selling price of products. This project analyzes a dataset containing product information and prices before and after the application of tariffs.

The project is developed as an interactive Python GUI-based Data Analytics application. It allows users to explore the dataset, perform data cleaning, analyze prices, generate different visualizations, compare products and categories, and understand the overall impact of tariffs.

The application provides a dashboard-style interface where different analytical features are organized into separate sections for easy navigation.

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### 3. Objective

The main objectives of the project are:

1. To analyze product prices before and after tariffs.
  2. To identify the change in product prices due to tariffs.
  3. To perform data cleaning and handle missing and duplicate values.
  4. To analyze products based on product type and brand.
  5. To calculate statistical measures such as mean, median, minimum, maximum and standard deviation.
  6. To create different graphical representations of the dataset.
  7. To provide an interactive GUI for exploring and analyzing the data.
  8. To provide a tariff simulator for estimating the effect of different tariff percentages on product prices.
  9. To generate useful observations and insights from the dataset.
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### 4. Dataset Description

The project uses a CSV dataset named **Tariff.csv**.

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## SUBJECT : PYTHON FOR DATA SCIENCE

The dataset contains information about products and their prices before and after tariffs.

### Dataset Columns

| Column                     | Description                                  |
|----------------------------|----------------------------------------------|
| <b>Serial Number</b>       | Unique serial number assigned to each record |
| <b>Product Name</b>        | Name of the product                          |
| <b>Product Type</b>        | Category or type of the product              |
| <b>Brand Name</b>          | Brand associated with the product            |
| <b>Price Before Tariff</b> | Product price before applying the tariff     |
| <b>Price After Tariff</b>  | Product price after applying the tariff      |

The dataset contains 3 numerical columns and 3 categorical/string columns.

### Numerical Columns

- Serial Number
- Price Before Tariff
- Price After Tariff

### Categorical Columns

- Product Name
- Product Type
- Brand Name

For demonstrating the data-cleaning process, a separate 100-row modified dataset is used. Missing values and duplicate records are intentionally introduced into this sample so that the cleaning operations can be demonstrated even though the original dataset is clean.

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## 5. Tools and Libraries Used

The project is developed using Python and several libraries.

1. **Tkinter** – Used to create the graphical user interface, including windows, buttons, frames, labels and menus.
2. **ttk** – Used for styled Tkinter widgets such as tables and scrollbars.
3. **Pandas** – Used for reading the CSV file and performing data cleaning, filtering, grouping and analysis.

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## SUBJECT : PYTHON FOR DATA SCIENCE

4. **NumPy** – Used for numerical calculations and statistical operations.
  5. **Matplotlib** – Used to create different types of graphs and charts.
  6. **Seaborn** – Used for statistical visualizations such as box plots.
  7. **BeautifulSoup** – Used for parsing and processing HTML content.
  8. **Pathlib** – Used for handling dataset and file paths.
  9. **FigureCanvasTkAgg** – Used to display Matplotlib graphs directly inside the Tkinter GUI.
  10. **Messagebox** – Used to display error, warning, confirmation and information messages in the GUI.
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## 6. Data Cleaning

Data cleaning is an important step before performing analysis because incorrect, missing or duplicate data can affect the results.

The original `Tariff.csv` dataset does not contain missing or duplicate values. Therefore, a separate 100-row dirty dataset was prepared for demonstrating the cleaning process.

The following operations are performed:

### a. Detecting Missing Values

The program checks each column for null or missing values using Pandas.

```
isnull().sum()
```

This helps identify which columns contain missing information.

### b. Handling Numerical Missing Values

Missing values in numerical columns are replaced using the mean value of the respective column.

For example:

Price Before Tariff → Mean of available prices

Price After Tariff → Mean of available prices

### c. Handling Categorical Missing Values

Missing values in text-based columns such as Product Name, Product Type and Brand Name are replaced with:

Unknown

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**d. Removing Duplicate Records**

Duplicate rows are detected and removed using:

`drop_duplicates()`

**e. Column Verification**

Column names are cleaned by removing unnecessary spaces, and numerical columns are converted into appropriate numeric data types.

After cleaning, the processed dataset can be saved as:

**Tariff\_cleaned.csv**

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## 7. Data Analysis

The project performs several statistical and categorical analyses on the tariff dataset.

### Statistical Analysis

The following measures are calculated:

- Mean
- Median
- Minimum
- Maximum
- Sum
- Standard deviation

These calculations are performed for:

- Price Before Tariff
- Price After Tariff
- Tariff Increase

### Tariff Increase

The difference between the prices is calculated as:

**Tariff Increase = Price After Tariff – Price Before Tariff**

The percentage impact can be calculated as:

**Tariff Impact % = ((Price After Tariff – Price Before Tariff) / Price Before Tariff) × 100**

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### Category Analysis

The program groups products according to Product Type to determine:

- Number of products in each category
- Average price after tariff
- Price distribution among categories

### Brand Analysis

Products can also be grouped according to Brand Name to examine:

- Number of products for each brand
  - Average price
  - Price differences between brands
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## 8. Data Visualizations

The project contains a separate Visualization section where different graphs can be generated to understand the dataset visually.

The application supports several types of visualizations:

### 1. Bar Chart

Used to compare numerical values between different categories, such as the average price of different product types.

### 2. Horizontal Bar Chart

Used to display category comparisons horizontally, making long category names easier to read.

### 3. Line Chart

Used to observe changes or trends in product prices.

### 4. Histogram

Used to show the frequency distribution of product prices.

### 5. Pie Chart

Used to show the proportion of different product types or categories in the dataset.

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### 6. Scatter Plot

Used to study the relationship between:

Price Before Tariff and Price After Tariff.

### 7. Box Plot

Used to examine the distribution, spread and possible outliers of prices across different product categories.

### 8. Custom Graph Generation

The application also provides a custom visualization option where the user can select:

- Dataset category
- Brand
- Product type
- Numerical variable
- Aggregation method
- Graph type

This allows users to generate graphs according to their own requirements instead of depending only on predefined graphs.

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## 9. Dashboard

The project provides an interactive Tariff Analytics Dashboard developed using Tkinter.

The dashboard is divided into different functional sections.

### Main Features

#### Dataset Explorer

Allows users to view and search the dataset.

#### Data Cleaning

Allows users to examine missing values and duplicate records and perform cleaning operations.

#### Analysis

Displays statistical calculations and category-based analysis.

#### Product Intelligence

Provides information about products, categories and brands.

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### Visualizations

Provides different graphs and a custom graph generation facility.

### Dataset Comparison

Allows comparison between different aspects of the dataset.

### Rankings

Displays products, categories or brands according to relevant calculated values.

### Smart Insights

Generates important observations from the dataset.

### Tariff Simulator

Allows the user to select a product and apply a hypothetical tariff percentage to estimate its new price.

### Dashboard Design

The GUI uses a modern dashboard layout with:

- Sidebar navigation
- Separate feature sections
- Cards for important statistics
- Graph panels
- Search and selection controls
- Scrollable content
- Interactive buttons
- Light/dark interface support
- Consistent colour palette
- Larger and bold typography

This structure makes the application easier to navigate compared with a single-screen data analysis program.

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## 10. Key Findings/Observations

The analysis of the dataset provides several useful observations:

1. Prices after tariff are higher than the corresponding prices before tariff for products affected by the tariff.
2. The difference between the two prices can be used to determine the absolute tariff impact on each product.
3. The percentage impact allows products with different original prices to be compared more effectively.
4. Different product types can have different average prices after the tariff.

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5. The number of products belonging to each product type can be identified using categorical analysis.
  6. Brand-wise analysis helps identify differences in average product prices between brands.
  7. The histogram provides information about the distribution of product prices.
  8. The scatter plot helps visualize the relationship between the original product price and its price after tariff.
  9. Box plots can help identify differences in price distribution and possible outliers among product categories.
  10. The tariff simulator demonstrates how changing the tariff percentage can affect the estimated final product price.
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## 11. Conclusion

The Tariff Impact Analysis on Product Prices project demonstrates how Python can be used to perform complete data analysis through an interactive graphical interface.

The project covers the complete data-analysis workflow, including data exploration, data cleaning, statistical analysis, visualization, comparison and insight generation. A separate modified dataset is also used to demonstrate how missing values and duplicate records can be handled.

The interactive dashboard makes the analysis easier to understand by presenting numerical results and visualizations in an organized manner. The custom visualization feature allows users to select their own variables and graph types, while the tariff simulator provides an interactive way to understand potential changes in product prices.

Overall, the project demonstrates the practical use of Python, Pandas, NumPy, Matplotlib, Seaborn and Tkinter in developing a complete data analytics application.